



Xiaomi Log Profile

White Paper

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Document Version History		
Version	Author(s)	Change Note
2024-11-08	Junkai Zhang & Guidong Zhao	Initial document draft
2024-12-05	Taoyuan Liu & Ye Luan	Clarify Mi-Log Curve domain, mirrored Mi-Log Curve and sensor-linear definition, add hardware encoding section
2024-12-17	Taoyuan Liu & Ye Luan	Expand the effective encoding range of Mi-Log functions, add explanations of related physical meanings, and update the parameters of codec functions



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1 Introduction

The Mi-Log profile employs a logarithmic curve and an extensive color gamut, making it a scene-referred encoding. The application of logarithmic encoding allows for a more efficient distribution of limited bits depth resources across varying exposure stops, thereby enhancing the video's dynamic range. Log captures are not designed for immediate viewing; they are intended to function as source clips for post-production processes.

2 Hardware Encoding

The Hardware Encoding Curves shown in Figure 1 in this section illustrate the Log video signal encoding used by Xiaomi SmartPhone - Flagship Series camera modules, but please note that for clarity, here we only display one full range hardware encoding curve of serval curves used in Xiaomi phone camera modules, and it is not for the specification intended for third-party implementation. For more information on third-party specifications, please refer to Section 3.

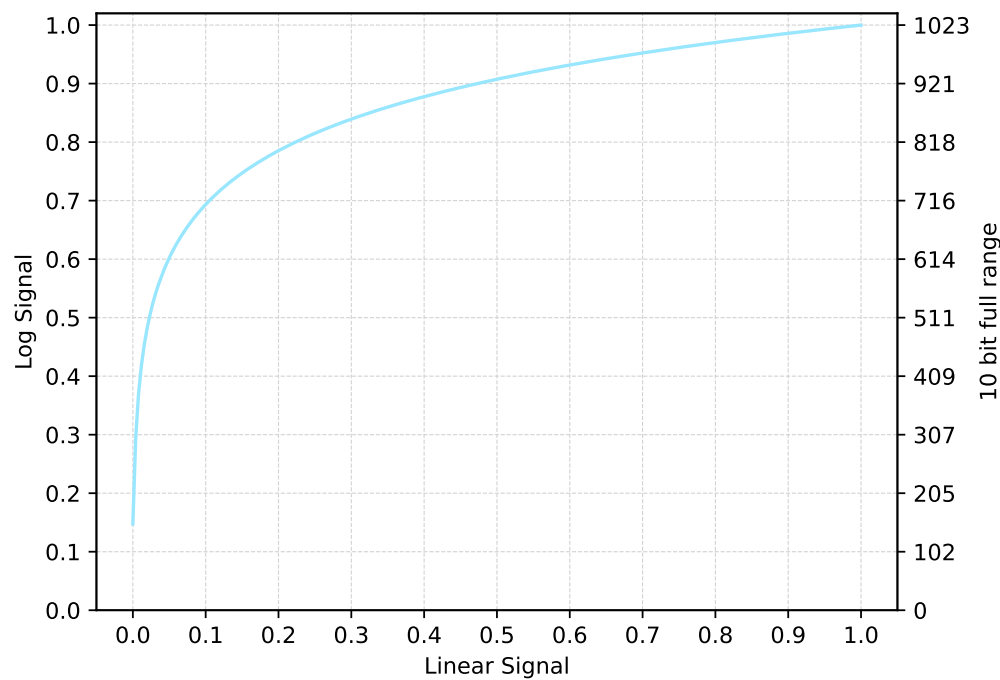


Figure 1: Hardware Encoding Curve for linear signal.



3 Transfer Function

The Mi-Log Curve serves as the transfer function in Mi-Log video encoding. It is a scene-referred logarithmic function defined by dedicated encoding and decoding functions. Notably, the Mi-Log Curve is tailored for 10-bit encoding, marking it as the minimum quantization level suitable for this logarithmic encoding scheme.

Figures 2 and 3 below illustrate the actual performance of the Mi-Log curve, where Figure 2 shows the relationship between stops relative to 18% gray and their corresponding brightness levels, and Figure 3 corresponds to the Log Curve output values, with the horizontal axis representing 18% gray scene reflection and the vertical axis representing the Log Curve output values.

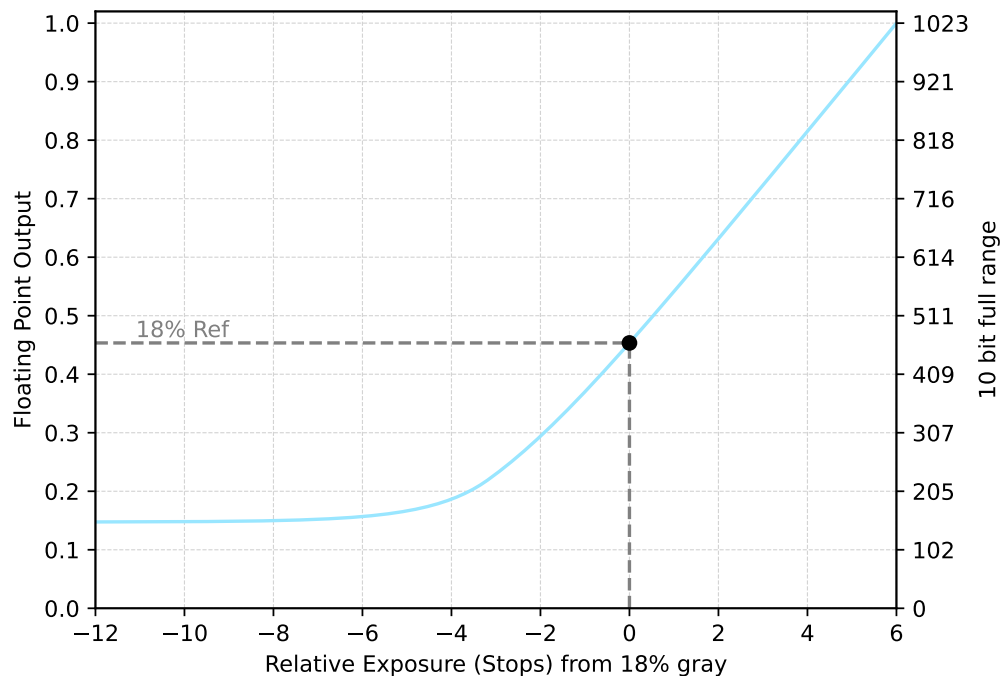


Figure 2: Xiaomi Log Curve for stops as normalized Float signal.

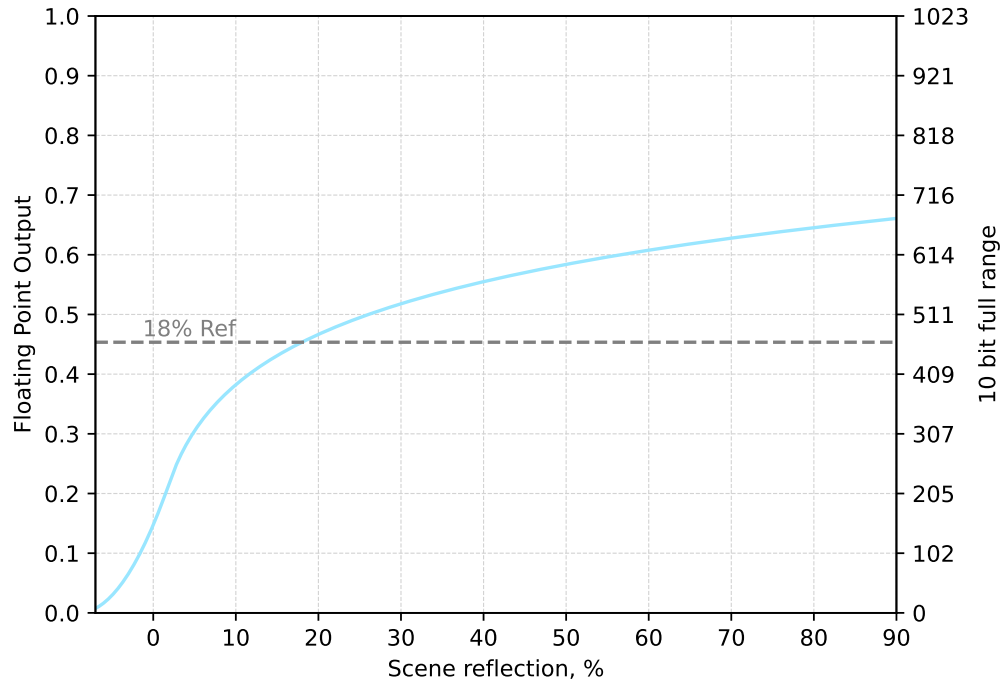


Figure 3: Xiaomi Log Curve for scene reflection as normalized Float signal

Scene reflection (%)	Encoded signal	
	Normalized float	10-bit integer, full range
0	0.14742742	150
18	0.45345968	463
90	0.66086763	676
1152	1.0	1023

Table 1: Xiaomi Log Curve's encoded signal with scene reflection

The max scene reflection 1152% found in the table 1 is derived by the following equation:

$$1152\% = 18\% \times 2^6 = \text{Scene}_{\text{Ref}} \times 2^{\text{Stop}_{\text{Max}}}$$

3.1 Encoding function

The encoding function, integrated within the camera system and presented herein for informational clarity, is designed for encoding relative scene linear values. It is formally defined as follows:

$$P = f(R) = \begin{cases} \gamma \log_2(R + \beta) + \delta, & \text{if } R \geq R_t \\ c(R - R_0)^2, & \text{if } R_0 \leq R < R_t \\ 0, & \text{if } R < R_0 \end{cases}$$

with

$$R_0 = -0.09023729$$

$$R_t = 0.01974185$$

$$c = 18.10531998$$

$$\gamma = 0.09271529$$

$$\beta = 0.01384578$$

$$\delta = 0.67291850$$

The overall encoding function is divided into two segments, delineated by the threshold R_t . Values exceeding this threshold are represented by a natural logarithmic relationship, while values below this threshold exhibit a quadratic relationship due to their higher growth rate.

3.2 Decoding function

The decoding function can be utilized on captured footage to restore the linear scene reflection signal. This p-decoding function is defined by the following equation:

$$R = f^{-1}(P) = \begin{cases} 2^{\frac{P-\delta}{\gamma}} - \beta, & \text{if } P \geq P_t \\ \sqrt{\frac{P}{c}} + R_0, & \text{if } 0 \leq P < P_t \\ R_0, & \text{if } P < 0 \end{cases}$$

here

$$P_t = c(R_t - R_0)^2$$

The captured pixel value P is using a floating point encoding normalized to the $[0,1]$ range.

4 Color Space

The color space defines the representation of colors. Mi Log employs RGB primaries and color space conversion matrices outlined in the ITU-R BT. 2020-2 specification. The white point is aligned with the D65 illuminant, and all chromaticity values are specified using the CIE 1931 2 Degree Standard Observer Colorimetry.

	CIE chromaticity coordinates	
	x	y
Red primary (R)	0.708	0.292
Green primary (G)	0.170	0.797
Blue primary (B)	0.131	0.046
White point (D65)	0.3127	0.3290

$$Y = 0.2627R + 0.6780G + 0.0593B$$

$$C_B = \frac{B - Y}{1.8814}$$

$$C_R = \frac{R - Y}{1.4746}$$